15.4.2 Types of polymerisation

The process of forming a very large, high molecular mass polymer from small structural units i.e., monomer is called polymerisation. Polymerisation occurs in the following two ways

i. Addition polymerisation or chain growth polymerisation
ii. Condensation polymerisation or step growth polymerisation

Addition polymerisation

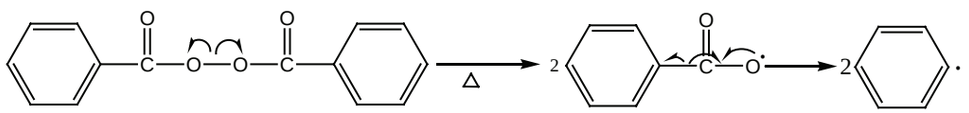
Many alkenes undergo polymerisation under suitable conditions. The chain growth mechanism involves the addition of the reactive end of the growing chain across the double bond of the monomer. The addition polymerisation can follow any of the following three mechanisms depending upon the reactive intermediate involved in the process.

i. Free radical polymerisation
ii. Cationic polymerisation
iii. Anionic polymerisation

Free radical polymerisation

When alkenes are heated with free radical initiator such as benzoyl peroxide, they undergo polymerisation reaction. For example styrene polymerises to polystyrene when it is heated to ionic with a peroxide initiator. The mechanism involves the following steps.

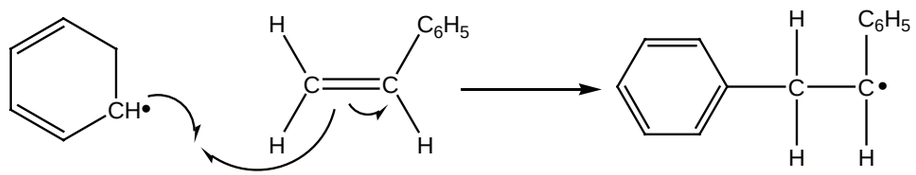
1. initiation – formation of free radical



benzoyl peroxide

phenyl free radical
+ $2CO_2$

2. Propagation step

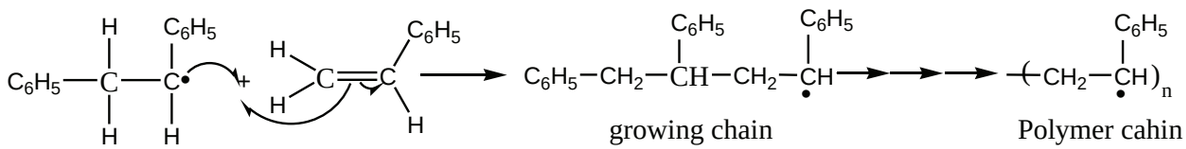


phenyl free radical

Styrene
(monomer)

Stabilised radical

The stabilized radical attacks another monomer molecule to give an elongated radical



monomer

growing chain

Polymer cahin

Chain growth will continue with the successive addition of several thousands of monomer units.

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